ATESTool

Installation and User Guide

Wouter Deleersnyder

November 11, 2026

Abstract

This guide explains how to install ATESTool from its public demonstration repository, verify the installation, and run the three examples. The exercises cover a single model run, reciprocal interaction between ATES and groundwater extraction, and sensitivity to ATES well layout. No prior knowledge of the package internals is required.

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1 What ATESTool does

ATESTool is a configuration-driven tool for modelling Aquifer Thermal Energy Storage (ATES) systems and their interaction with groundwater extraction. It creates an unstructured mesh, builds coupled MODFLOW 6 groundwater-flow (GWF) and energy-transport (GWE) models through FloPy, runs the model, and writes processed results.

The public repository is available at:

https://github.com/WouterDls/ATESTool-demo

The repository has three main directories:

`ATESTool/` Python source code.

`docs/` Technical and API documentation.

`examples/` The shared example configuration and the three runnable examples.

2 Installation

Use Python 3.12 for the examples. Conda installs it inside the ATESTool environment. Choose your platform: Windows, Linux, or macOS. Linux and macOS share the same commands once Conda and Git are ready. See Appendix A for alternative installation options.

2.1 Windows setup

If needed, follow the Windows Miniconda instructions. Open **Anaconda Prompt** from the Start menu. Run `git -version`. If Git is missing, install Git for Windows, reopen Anaconda Prompt, and repeat the check.

1. Create the Python environment. Enter each command separately. If Conda asks whether to proceed, enter `y`. Check that the last line reports `Python 3.12.x` before continuing.

```
conda create -n atestool python=3.12 pip
conda activate atestool
python --version
```

2. Clone the repository. In the terminal, go to the folder where you want to keep ATESTool. Use `mkdir models` to create a folder, `cd models` to enter it, and `cd ..` to go up one level. Use `dir` to list the current folder. Press Up Arrow to recall a previous command.

```
git clone https://github.com/WouterDls/ATESTool-demo.git
cd ATESTool-demo
```

3. Install ATESTool. The `mesh` extra installs Triangle for the refined meshes used in the exercises. Triangle has commercial-use restrictions. Review its licence before commercial use.

```
python -m pip install ".[mesh]"
ATESTool --help
```

4. Install and check MODFLOW 6. FloPy's downloader puts MODFLOW 6 in the active environment. The examples use version 6.6.1. Check that `mf6 -version` reports this version.

```
get-modflow --repo modflow6 --release-id 6.6.1 :python
mf6 --version
```

5. Run the first example. This simulates five years and shows progress in the terminal while MODFLOW runs.

```
ATESTool examples/single_run.yaml --workspace examples/output/e1_single_run --report
```

Open `examples/output/e1_single_run/report.html` in a browser. When opening a new terminal, run `conda activate atestool` and return to the repository folder before running the next example.

2.2 Linux setup

If needed, follow the Linux Miniconda instructions. Initialize Conda for your shell when prompted and open a new terminal. Run `git -version`. If Git is missing, run `sudo apt install git` on Debian or Ubuntu, or follow the Git instructions for your distribution.

2.2.1 Commands for Linux and macOS

1. Create the Python environment. Enter each command separately. If Conda asks whether to proceed, enter `y`. Check that the last line reports `Python 3.12.x` before continuing.

```
conda create -n atestool python=3.12 pip
conda activate atestool
python --version
```

2. Clone the repository. In the terminal, go to the folder where you want to keep ATESTool. Use `mkdir models` to create a folder, `cd models` to enter it, and `cd ..` to go up one level. Use `ls` to list the current folder. Press Up Arrow to recall a previous command.

```
git clone https://github.com/WouterDls/ATESTool-demo.git
cd ATESTool-demo
```

3. Install ATESTool. The `mesh` extra installs Triangle for the refined meshes used in the exercises. Triangle has commercial-use restrictions. Review its licence before commercial use.

```
python -m pip install ".[mesh]"
ATESTool --help
```

4. Install and check MODFLOW 6. FloPy's downloader puts MODFLOW 6 in the active environment. The examples use version 6.6.1. Check that `mf6 -version` reports this version.

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get-modflow --repo modflow6 --release-id 6.6.1 :python
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Open `examples/output/e1_single_run/report.html` in a browser. When opening a new terminal, run `conda activate atestool` and return to the repository folder before running the next example.

2.3 macOS setup

If needed, follow the macOS Conda instructions and open a new Terminal window. Run `git -version`. If Git is missing, run `xcode-select -install` and repeat the check. Then follow Steps 1–5 under Commands for Linux and macOS.

3 Exercise 1: Run and inspect one model

This exercise introduces the YAML-first workflow.

3.1 Learning objectives

After this exercise, you should be able to run a YAML configuration, identify the generated files, and use the HTML report to check the model input and principal results.

3.2 Task 1: inspect the configuration

Open `examples/single_run.yaml`. Identify the following blocks before running it:

`mesh` Domain dimensions, background cell size, and refinement rings.

`model.aquifer` Hydraulic and thermal properties of the aquifer layer.

`model.chd_heads` Regional hydraulic-head boundary conditions.

`model.doublets` ATES well locations, flow rate, and injection temperatures.

`model.extraction_well` Position and pumping rate of the extraction well.

`model.simulation` Duration, seasonal length, and time stepping.

`analysis` Requested output variables and reporting times.

Coordinates are metres, time is days, flow is cubic metres per day, and temperature is degrees Celsius.

3.3 Task 2: run the model and create a report

If you already ran this example during installation, use those results and continue to Task 3. Otherwise, run this command from the repository root.

```
ATESTool examples/single_run.yaml --workspace examples/output/e1_single_run --report
```

Using a separate workspace is important: it prevents the files from one configuration from being mixed with those of another. The `-report` option creates a compact HTML report from the completed run without launching extra MODFLOW simulations.

3.4 Task 3: inspect the output

Open `examples/output/e1_single_run/report.html`. Locate the model layout, regional flow, well-centred drawdown, temperature results, extraction-well time series, energy balance, and recovery efficiency. Then find these reusable data files in the same folder:

`timeseries.csv` Extraction-well temperature and head through time.

`energy.csv` Seasonal energy injected, extracted, and recovered.

`recovery.csv` Recovery efficiencies arranged by ATES well.

`stats.json` Summary statistics for the run.

Exercise questions Record the simulated duration, the direction of regional groundwater flow, the maximum drawdown close to the extraction well, and the final extraction temperature. Explain which figure or table you used for each answer.

4 Exercise 2: Case 1—reciprocal interaction

4.1 Question

How does a balanced ATES system change the temperature of extracted groundwater, and how does groundwater extraction in turn change ATES recovery? Answering both parts requires three matched physical states.

4.2 Task 1: run the three states

Run each YAML file independently from the repository root:

```
ATESTool examples/case1/combined.yaml --workspace examples/output/case1/combined --
    report
ATESTool examples/case1/extraction_only.yaml --workspace examples/output/case1/
    extraction_only --report
ATESTool examples/case1/ates_only.yaml --workspace examples/output/case1/ates_only --
    report
```

The states are:

`combined` ATES and groundwater extraction are active.

`extraction_only` Extraction is active and ATES flow is zero.

`ates_only` ATES is active and groundwater extraction is zero.

All three configurations use the same mesh settings and mesh-only refinement points, so their results can be compared cell for cell. Each workspace contains its own HTML report and CSV outputs. Inspect the three reports before aggregating any data.

4.3 Task 2: construct the comparison in Excel

Open `examples/case1/case1_analysis_template.xlsx` and read its `Guide` sheet. Paste *values only*, including the header row, at cell A1 of each yellow source sheet:

Generated file	Excel source sheet
`combined/timeseries.csv`	`Combined time`
`extraction_only/timeseries.csv`	`Extraction time`
`combined/energy.csv`	`Combined energy`
`ates_only/energy.csv`	`ATES energy`

Do not paste into `Analysis`, `Recovery`, or `Summary`; those sheets contain formulas. Save your completed workbook under a new name.

4.4 Task 3: interpret the reciprocal effects

Use the `Analysis` and `Summary` sheets to determine the maximum ATES-induced warming and cooling at the extraction well. Use `Recovery` to compare mean ATES recovery with and without groundwater extraction.

Exercise questions Which comparison isolates the effect of ATES on extraction temperature? Which comparison isolates the effect of extraction on ATES recovery? Report the magnitude and sign of both effects and explain what the sign means physically.

After completing the exercise, compare your workbook with `examples/case1/case1_analysis_solution.xlsx`. The solution is a checking aid; do not use it as the source for your own analysis.

5 Exercise 3: Case 2—layout sensitivity

5.1 Question

How do distance, orientation, and warm/cold-well ordering change the interference signal when aquifer properties and operating rates remain fixed?

5.2 Task 1: run the reference and four layouts

```
ATESTool examples/case2/without_ates.yaml --workspace examples/output/case2/
    without_ates --report
ATESTool examples/case2/near_upstream.yaml --workspace examples/output/case2/
    near_upstream --report
ATESTool examples/case2/far_upstream.yaml --workspace examples/output/case2/
    far_upstream --report
ATESTool examples/case2/transverse.yaml --workspace examples/output/case2/transverse --
    report
ATESTool examples/case2/reversed_longitudinal.yaml --workspace examples/output/case2/
    reversed_longitudinal --report
```

`without_ates` Common groundwater-extraction reference.

`near_upstream` Longitudinal doublet close to the extraction well and upstream under the imposed regional flow.

`far_upstream` The same layout farther upstream.

`transverse` Doublet perpendicular to regional flow.

`reversed_longitudinal` Longitudinal warm/cold ordering reversed.

Open every `report.html`. Confirm that the well layouts match their descriptions and compare the individual extraction-temperature histories.

5.3 Task 2: construct the comparison in Excel

Open `examples/case2/case2_analysis_template.xlsx`. For each run, copy its complete `timeseries.csv`, including the header, and paste values only at A1 of the matching yellow sheet: `Without ATES`, `Near time`, `Far time`, `Transverse time`, or `Reversed time`. The formulas subtract the common reference and update the chart and summary automatically.

5.4 Task 3: interpret the layout effects

Use the calculated histories and summary metrics to answer:

1. Does moving the doublet farther away reduce the maximum interference signal?
2. Does the transverse layout cause warm and cold signals to partly cancel?
3. Does reversing the longitudinal ordering change the sign of the response?
4. Which layout has the largest absolute temperature change, and which has the largest final-year mean change?
5. Why is comparison with `without_ates` preferable to comparison with the ambient temperature alone?

Save your completed workbook under a new name. Then compare it with `examples/case2/case2_analysis_solution.xlsx`. Use any disagreement to check that the complete CSV files were pasted into the correct sheets and that the header rows begin at A1.

6 Recommended workflow for new cases

Use the following sequence when adapting the examples:

1. Copy `examples/base.yaml`
2. Replace the aquifer properties.
3. Set the regional boundary heads and verify the implied flow direction.
4. Define extraction and ATES well coordinates and rates.
5. Start with a shorter simulation for setup checks, and run it in its own workspace:

```
ATESTool my_case.yaml --workspace output/my_case
```

6. Optional: Review the mesh-quality, Péclet, thermal-resolution, and Courant numbers logged during the run. For a full verification bundle, add `-diagnostic`; results are written to `<workspace>/diagnostic/`, with an overview in `summary.txt`. This costs roughly twelve baseline runs.
7. Optional: Refine space and time where the decision variable is sensitive.
8. Analyse complete time series as well as summary metrics.

7 Troubleshooting

`ModuleNotFoundError: ATESTool` Activate the ATESTool environment and rerun `python -m pip install ".[mesh]"` from the repository root.

MODFLOW executable not found Activate the ATESTool environment and repeat Step 4. For an existing executable, see Appendix A.

GWE package or input is unsupported Use MODFLOW 6 version 6.6.1 to reproduce the supplied examples.

Triangle is unavailable Install `triangle` or reinstall with `.[mesh]`. Review Triangle's licence before commercial use.

The simulation is slow Reduce the number of simulated years while checking setup, but restore the intended duration for analysis. Mesh refinement, time steps, and diagnostic studies all increase runtime.

Mesh, Péclet, thermal-resolution, or Courant warnings Treat these as accuracy advisories. Tighten spatial refinement or time stepping and compare the decision variable between successive levels.

An output directory already exists Choose a new workspace or deliberately remove old generated output before rerunning. Keeping separate workspaces prevents accidental mixing of results.

8 Further documentation

The repository's `README.md` provides a concise overview. More detailed configuration, meshing, CLI, and API information is in `docs/`. To build the technical HTML documentation:

```
python -m pip install -e ".[docs]"
sphinx-build -b html docs docs/_build/html
```

A Alternative installation options

A.1 Use an existing Python 3.12 installation

If you need a standalone interpreter, follow the Python 3.12 setup instructions for your platform. Clone or extract the repository and change into its folder. Use the explicit 3.12 interpreter to create a virtual environment.

Windows Command Prompt

```
py -3.12 --version
py -3.12 -m venv .venv
.venv\Scripts\activate.bat
python --version
```

Linux or macOS (Bash or Zsh)

```
python3.12 --version
python3.12 -m venv .venv
source .venv/bin/activate
python --version
```

Check for `Python 3.12.x`, then continue with Step 3 for your platform. If an existing virtual environment uses a different Python version, create a new one with Python 3.12.

A.2 Download the repository as a ZIP file

On GitHub, choose **Code** and then **Download ZIP**. Extract the archive and use `cd` to enter the extracted folder in place of Step 2.

A.3 Editable installs and optional packages

For source development, install in editable mode. Add the development and documentation extras if you need the type checker and documentation tools.

```
python -m pip install -e ".[mesh]"
python -m pip install -e ".[mesh,dev,docs]"
```

To install without Triangle, run `python -m pip install .`. The SciPy fallback produces meshes without per-well refinement.

A.4 Install MODFLOW 6 manually

Download the build for your operating system and CPU from the MODFLOW 6.6.1 release page. ATESTool requires a version newer than 6.5.0. Use the ARM64 build on Apple Silicon. Either add the binary directory to `PATH` and run `mf6 -version`, or point ATESTool directly at the executable and check that file's version.

Windows Command Prompt

```
set "MF6_BIN=C:\path\to\mf6.exe"
"%MF6_BIN%" --version
```

Linux or macOS

```
export MF6_BIN="/path/to/mf6"
"$MF6_BIN" --version
```